

Labwork 2: Linear Regression

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Deep Learning – USTH

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1 Implementation

```
def read_csv(path):
    xs = []
    ys = []
    f = open(path, "r")
    for line in f:
        line = line.strip()
        if line == "":
            continue
        parts = line.split(",")
        xs.append(float(parts[0]))
        ys.append(float(parts[1]))
    f.close()
    return xs, ys

def predict(x, w0, w1):
    return w1 * x + w0

def loss(w0, w1):
    s = 0.0
    for i in range(N):
        e = predict(xs[i], w0, w1) - ys[i]
        s = s + e * e
    return 0.5 * s / N

def train(w0, w1, r, iters):
    for it in range(iters):
        for i in range(N):
            e = w1 * xs[i] + w0 - ys[i]
            w0 = w0 - r * e
            w1 = w1 - r * xs[i] * e
        print(f"Iteration {it+1}: w0 = {w0:.4f}, "
              f"w1 = {w1:.4f}, J = {loss(w0, w1):.4f}")
    return w0, w1
```

The training loop walks through all five data points each iteration and updates w_0 and w_1 after every sample.

2 Results

The model is trained from initialization $w_0 = 0$, $w_1 = 1$ with learning rate $r = 10^{-4}$ for 100 iterations. The first ten iterations are shown below.

iter	w_0	w_1	$J(w_0, w_1)$
1	0.0217	1.7763	443.9450
2	0.0312	1.9189	358.4757
3	0.0385	1.9450	348.0546
4	0.0453	1.9498	346.2428
5	0.0521	1.9506	345.8382
6	0.0589	1.9506	345.6862
7	0.0657	1.9506	345.5806
8	0.0725	1.9505	345.4835
9	0.0792	1.9504	345.3880
10	0.0860	1.9503	345.2928

After 100 iterations the model converges to

$$w_0 = 0.6915, \quad w_1 = 1.9416, \quad J = 336.8393,$$

giving the fit shown in Figure 1.

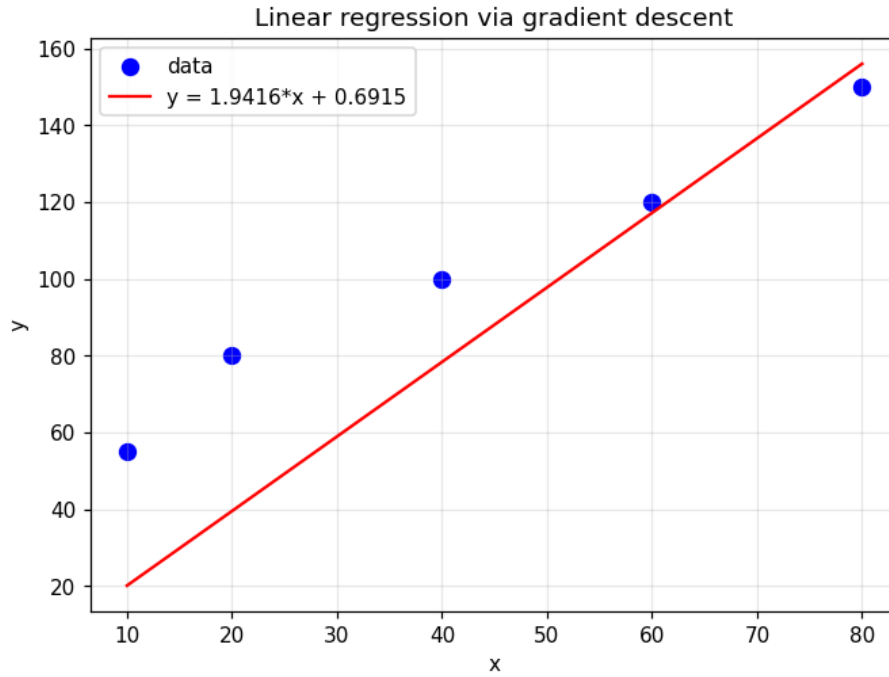


Figure 1: Fitted line $y = 1.9416x + 0.6915$ obtained from 100 iterations.